

SUMMARY REPORT

OJEYINKA, MARY INIOLUWA

Generative Artificial Intelligence Internship Program – AnalystLab Africa

1.0 Key Findings from Exploratory Data Analysis (EDA)

Analysis of the Titanic dataset revealed several important patterns. Female passengers exhibited a significantly higher survival rate than male passengers, indicating that gender played a major role in survival outcomes. Passengers travelling in first class were more likely to survive than those in second and third classes, suggesting that socioeconomic status influenced access to safety measures. The majority of passengers were between 20 and 40 years of age, and most passengers travelled in third class. Additionally, higher fare values were generally associated with higher survival rates, reflecting the relationship between ticket class and survival probability.

2.0 Challenges Encountered

One of the primary challenges encountered during this assignment was adapting to the notebook-based workflow and organizing analyses in a clear and professional manner. Understanding how to properly handle missing values without affecting the integrity of the dataset required careful consideration. Another challenge involved interpreting visualizations and translating graphical patterns into meaningful insights rather than simply generating charts. During the NLP task, preprocessing textual data required understanding multiple cleaning steps and their importance in preparing data for machine learning applications.

3.0 Lessons Learned During Weeks 1–2

The first two weeks of the internship reinforced the importance of data preparation as a foundation for Artificial Intelligence and Machine Learning projects. I learned that high-quality

analysis depends on well-cleaned and properly structured data. Through Exploratory Data Analysis, I developed a better understanding of how visualizations can be used to identify trends, relationships, and patterns within datasets. The NLP exercises demonstrated the importance of text preprocessing techniques such as lowercasing, stopword removal, and tokenization in transforming raw text into a format suitable for analysis. Overall, these activities strengthened my practical skills in Python, data analysis, data visualization, and introductory Natural Language Processing.